Web Application Security Testing

Cyber Security & Ethical Hacking

(Level 2)

1. Lab Setup :

- Configured a secure local lab environment using Kali Linux virtual machine.
- Prepared tools for web application vulnerability scanning and analysis.

2. Tools Used :

- Kali Linux
- Nikto (Web Server Scanner)
- Burp Suite (Proxy / Web App Testing)

3. Testing Methodology & Execution :

- **Step 1:** Verified security testing tool configurations in the Kali Linux environment.
- **Step 2:** Executed Nikto web vulnerability scanner to identify potential web server misconfigurations and vulnerabilities:

```
nikto -h localhost
```

- **Step 3:** Analyzed security misconfigurations and simulated web testing workflows using proxy analysis (Burp Suite).

4. Screenshots / Output :





5. Mitigation Strategies & Recommendations :

- Regularly update web servers and applications to patch known vulnerabilities.
- Implement strict input validation to prevent SQL Injection and XSS attacks.
- Configure security headers and disable default administrative consoles.

6. Conclusion :

. Successfully performed web application security testing principles in a controlled local lab environment using Nikto and Burp Suite, documenting security risks and appropriate mitigation strategies.